

The Cell Culture Industrialist

Dr. Vishal G. Warke, A visionary to make Bharat *Atmanirbhar* in Biotechnology

Aligned with the Hon'ble Prime Minister Modi sir's vision of Atmanirbhar Bharat, Dr. Vishal's scale-up strategy directly addresses the national priorities and positions HiMedia as a biomanufacturing partner of choice. This, in many ways, is the operational mantra that will help propel Bharat towards its Amritkaal aspirations.

When Dr. Vishal Warke joined HiMedia Laboratories in 2003, the a company founded in 1974 by Dr. Gangadhar M. Warke, Mrs. Saroj G. Warke and Shri. Vishnu Warke was already a highly respected producer of microbiology media serving researchers globally. Two decades later, under the stewardship of Dr. Vishal G. Warke & his wife Dr. Priti V. Warke in cell biology, Dr. Rajas V. Warke & Dr. Kavita Khadke in Molecular biology and Dr. Rahul G. Warke in Microbiology, HiMedia has evolved into a bioscience conglomerate operating across 150 countries with 30,000+ SKUs spanning pharmaceuticals, food and beverage, cosmetics, agriculture, and biopharma sectors.

The transformation in the Cell biology reflects Dr. Vishal's systematic expansion beyond commodity products into high-margin specialty applications. His MBBS from Mumbai University and PhD in Cell Biology and Molecular Genetics from the University of Maryland positioned him uniquely understanding both clinical needs and molecular mechanisms that most business-focused successors lack.

"True entrepreneurial success lies in building systems that outlast individual achievements—creating value that ripples through industries, communities, and generations," Dr. Warke explains, by articulating the institutional-building philosophy distinguishing HiMedia from purely transactional bioscience suppliers.

The diversification strategy reflects Dr. Warke's spirit of calculated risk-taking. He expanded the organization by establishing Animal Tissue Culture, Plant Tissue Culture, and Hydroponics divisions, while pioneering the development of Chemically Defined, Serum-Free Media for biosimilar manufacturing, thus creating essential infrastructure that supports India's emergence as the "Pharmacy of the World." His serum-free media for BHK-21, Vero, and MDCK cells now serve vaccine manufacturers globally, while CHO cell media supports monoclonal antibody production.

However, it is HiMedia's investment in cultivated meat that most strongly reflects Dr. Warke's strategic foresight. While others waited for market validation, he chose to invest early in cGMP-certified, cost-effective facilities dedicated to the large-scale production of animal-free media formulations. "We don't just manufacture pharmaceuticals; we create possibilities for global health and economic progress," he says, positioning HiMedia as a key infrastructure provider for the global protein transition.

The COVID-19 response demonstrated operational agility. When India's government urgently needed Viral Transport Medium kits, Dr. Warke and Dr. Priti ramped production from 0.5 million test kits annually to 1 million daily—a 730x scale-up that strengthened national diagnostic capacity during crisis.

Research investment validates the innovation mandate. Dr. Warke allocates 16% of profits to R&D, funding acquisitions like the Orbitrap Fusion Tribrid Mass Spectrometer advancing metabolomics and proteomics capabilities. With over 20 peer-reviewed publications in leading journals such as the Journal of Cell Biology and the Journal of Biological Chemistry, he has established strong scientific credibility that extends far beyond commercial operations.

Beyond HiMedia's operations, Dr. Warke mentors biotech startups including Next Big Innovation Labs (3D bioprinting), KrishiTantra



Dr. Vishal Warke
Director of Cell Biology and Hydroponics at HiMedia Laboratories Pvt. Ltd

(precision agriculture), and Nuorentaa (personalized healthcare)—socializing expertise that accelerates India's bioeconomy development. His collaboration in establishing the Wadhvani Research Centre for Bioengineering at IIT Bombay demonstrates ecosystem-building extending beyond shareholder value.

The success of HiMedia's investment in cultivated meat media manufacturing infrastructure depends on regulatory approvals and how quickly consumers accept the technology. Aligned with the Hon'ble Prime Minister's vision of Atmanirbhar Bharat and the recently approved BIOE³ policy, Dr. Vishal has architected a state-of-the-art, cGMP- and USFDA-compliant culture media manufacturing facility with production capabilities spanning from 100 grams to 100 tons per day for both catalogue and fully customized media. This scalable, high-throughput infrastructure is strategically designed to serve the capacity needs of CDMOs, CROs and CMOs across the pharmaceutical, food, biopharma and vaccine sectors in India and globally. HiMedia's operations are further strengthened by ISO 9001, ISO 13485, ISO 22000 and kosher certifications, underscoring a robust quality and compliance framework. Under the DBT-BIRAC biofoundry and biomanufacturing schemes, success is defined by achievements at higher Technology Readiness Levels (TRLs), dependable scale-up capabilities, and sustained high-capacity manufacturing performance.